Signal-detection foundations of the value-directed-attention model

A derivation note for the rebuilt model section

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Abstract

This note derives, from first principles, the per-location hit-rate and false-alarm-rate expressions that the rebuilt model inherits from the Herman Lab manuscript, and states the modelling conventions that make them well-posed. The goal is to make every symbol in Equation (4) traceable to an explicit probabilistic statement about an observer's decision, and to localise the single place where the per-location independence assumption (A1) enters the N-location reward, which is the seam the rebuild later cuts along to introduce the decorrelation channel.

1 The task and the decision

An observer monitors $N \geq 2$ spatial locations (in the cued change-detection paradigm, N oriented patches). On each trial at most one location changes. The observer cannot read out the world directly; at each location it has access only to a scalar *decision variable* X_i — a noisy internal summary of the evidence that location i changed. The observer issues a separate binary judgement at each location, "change" or "no change." The quantities of interest are the two conditional response rates at each location: the *hit rate* HR_i (the probability of reporting "change" when location i truly changed) and the *false-alarm rate* FAR_i (the probability of reporting "change" when it did not).

We work throughout with the *normative* observer: rather than modelling any particular subject's idiosyncratic policy, we ask what an observer that has already tuned its sensitivities and criteria to maximise expected reward would do. The signal-detection layer below is the measurement model on which that optimisation is later performed.

2 The per-location signal-detection model

Distributions. Adopt the equal-variance Gaussian model with unit variance ($\sigma = 1$). At location i, the decision variable is drawn from one of two distributions according to whether a change occurred:

$$\text{no change } (H_0):\ X_i \sim \mathcal{N}\Big(-\tfrac{d'_i}{2}, 1\Big), \qquad \text{change } (H_1):\ X_i \sim \mathcal{N}\Big(+\tfrac{d'_i}{2}, 1\Big). \tag{1}$$

The two means are placed symmetrically about 0, so that their separation is exactly d'_i. This is the definition of sensitivity:

$$d'_i := \frac{\mathbb{E}[X_i \mid H_1] - \mathbb{E}[X_i \mid H_0]}{\sigma} = \tfrac{d'_i}{2} - \Big(-\tfrac{d'_i}{2}\Big), \qquad \sigma = 1. \tag{2}$$

Larger d'_i pushes the two distributions apart and makes change more discriminable from no-change.

Decision rule. The observer reports "change" at location i when the evidence exceeds a threshold, the *criterion* $c_i \in \mathbb{R}$:

$$\text{respond "change" at } i \iff X_i > c_i. \tag{3}$$

Because c_i is measured on the same centred axis as (1), $c_i = 0$ is the unbiased criterion (it cuts the two distributions symmetrically); $c_i > 0$ is conservative (fewer "change" responses), $c_i < 0$ liberal.

Target. Both HR_i and FAR_i are the *same* event under (3) — "X_i exceeds c_i" — evaluated under the two hypotheses of (1). They are therefore tail areas of Gaussians, computed with the standard-normal cumulative distribution function $\Phi(z) = \Pr(Z \le z)$, $Z \sim \mathcal{N}(0,1)$.

3 Deriving the hit and false-alarm rates

We use two elementary facts, stated as lemmas so the derivation has no hidden steps.

Lemma 1 (Standardisation)**.** *If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $Z := (X-\mu)/\sigma \sim \mathcal{N}(0,1)$, and for any $t \in \mathbb{R}$, $\Pr(X > t) = \Pr(Z > (t-\mu)/\sigma)$.*

Lemma 2 (Tail symmetry of the standard normal)**.** *For all $z \in \mathbb{R}$, $1 - \Phi(z) = \Phi(-z)$, because the standard-normal density is even about 0.*

Proposition 1 (Per-location SDT rates)**.** *Under the model (1)–(3) with $\sigma = 1$,*

$$\mathrm{HR}_i = \Phi\Big(\tfrac{d'_i}{2} - c_i\Big), \qquad \mathrm{FAR}_i = \Phi\Big(-\tfrac{d'_i}{2} - c_i\Big). \tag{4}$$

Proof. Hit rate. Condition on H_1, so $X_i \sim \mathcal{N}(d'_i/2, 1)$. Apply Lemma 1 with $\mu = d'_i/2$, $\sigma = 1$, $t = c_i$, and then Lemma 2:

$$\mathrm{HR}_i = \Pr(X_i > c_i \mid H_1) = \Pr\big(Z > c_i - \tfrac{d'_i}{2}\big) = 1 - \Phi\big(c_i - \tfrac{d'_i}{2}\big) = \Phi\big(\tfrac{d'_i}{2} - c_i\big). \tag{5}$$

False-alarm rate. Condition on H_0, so $X_i \sim \mathcal{N}(-d'_i/2, 1)$. Identically,

$$\mathrm{FAR}_i = \Pr(X_i > c_i \mid H_0) = \Pr\big(Z > c_i + \tfrac{d'_i}{2}\big) = 1 - \Phi\big(c_i + \tfrac{d'_i}{2}\big) = \Phi\big(-\tfrac{d'_i}{2} - c_i\big). \tag{6}$$

These are (4). $\square$

4 Conventions and what they buy

Remark 1 (The centring is a choice, not a law). Placing the means at $\mp d'_i/2$ fixes the origin of the decision axis at the midpoint of the two distributions. Any other origin gives an algebraically equivalent model. For instance, putting noise at 0 and signal at d'_i and measuring a criterion c'_i from the noise mean yields $\mathrm{HR}_i = \Phi(d'_i - c'_i)$, $\mathrm{FAR}_i = \Phi(-c'_i)$; the substitution $c_i = c'_i - d'_i/2$ recovers (4). The centred convention is preferred only because it makes $c_i = 0$ the bias-free criterion, so the sign of c_i reads directly as conservative/liberal.

Remark 2 (Why a threshold on X_i is general). The rule (3) is a simple threshold on the raw decision variable, not on a likelihood ratio. Under the equal-variance Gaussian model the likelihood ratio $\Lambda(x) = \varphi(x - d'_i/2)/\varphi(x + d'_i/2) = \exp(d'_i x)$ is strictly increasing in x, so "$\Lambda(X_i) > \tau$" is equivalent to "$X_i > c_i$" for a suitable c_i. By the Neyman–Pearson lemma the threshold rule is therefore the optimal test, and (3) loses no generality. This equivalence relies on equal variances: with $\sigma_0 \neq \sigma_1$ the likelihood ratio is non-monotone, the optimal region is an interval rather than a half-line, and (4) no longer holds. Equal variance is thus a load-bearing modelling assumption, not a cosmetic one.

5 From one location to N: the no-false-alarm term

The expected reward couples locations only through the no-change trials, where the observer is penalised for false-alarming *anywhere*. Define the per-location correct-rejection probability

$$\mathrm{CR}_i := 1 - \mathrm{FAR}_i = 1 - \Phi\Big(-\tfrac{d_i'}{2} - c_i\Big) = \Phi\Big(\tfrac{d_i'}{2} + c_i\Big) =: \Phi(b_i), \qquad b_i := c_i + \tfrac{d_i'}{2}, \tag{7}$$

using Lemma 2 again; b_i is the z-score the no-change decision variable must stay below to avoid a false alarm. The probability that *no* location false-alarms on a no-change trial is then

$$P_{\text{no-fa}} \;=\; \Pr\Bigg(\bigcap_i \{X_i \le c_i\} \,\Bigg|\, H_0 \text{ at every location}\Bigg). \tag{8}$$

This is the *only* term in the expected reward that involves more than one location at once: on a change trial the change is at a single location, so the change-trial reward is linear in the marginal hit rates and contains no cross-location product.

Where assumption A1 enters. If the per-location decision variables are independent (assumption A1), the joint event (8) factorises into a product of marginals, and with homogeneous uncued allocation (d_u', c_u shared across the $N-1$ uncued locations) it collapses to

$$P^{\text{indep}}_{\text{no-fa}} \;=\; \prod_i (1 - \mathrm{FAR}_i) \;=\; \Phi(b_c)\,\Phi(b_u)^{N-1}. \tag{9}$$

Equation (9) is the sole appearance of A1 in the model. The rebuilt manuscript replaces it with the exact equicorrelated-Gaussian orthant probability: writing the no-change variables in one-factor form $X_i = \mu_i + \sqrt{\rho}\,Z + \sqrt{1-\rho}\,\varepsilon_i$ with $Z, \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ and $\mu_i = -d_i'/2$, the variables are conditionally independent given Z, so multiplying the conditional correct-rejection probabilities and integrating out Z gives

$$P_{\text{no-fa}}(\rho) \;=\; \int_{-\infty}^{\infty} \Phi\bigg(\frac{b_c - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\bigg)\,\Phi\bigg(\frac{b_u - \sqrt{\rho}\,z}{\sqrt{1-\rho}}\bigg)^{N-1} \varphi(z)\,dz, \qquad \rho \in [0,1). \tag{10}$$

At $\rho = 0$ the integrand is constant in z and (10) reduces to (9); this is the recovery contract that pins the extended model to the inherited one. Equation (10) requires no multivariate-normal CDF — equicorrelation collapses the N-dimensional integral to a single dimension — and is the formal content of the rebuild's "decorrelation channel."

6 The two knobs and the question

Equations (4) expose the observer's two per-location controls. *Sensitivity* d_i' is set by attention through the asymmetric transfer $d_c' = d'_{\max} f(\alpha_c; \beta(r))$, $d_u' = d'_{\max} f(\alpha_u; \gamma(r))$, where the benefit/cost ratio r governs how much sensitivity is gained at the cued focus versus lost at the periphery. The *criterion* c_i sets the bias independently of sensitivity. Adapting to a valuable cue can in principle be done by sharpening d_c' (moving attention) or by lowering c_c (shifting the criterion), and the central question of the paper — when does value-directed attention matter? — is precisely which of these the reward-maximising observer relies on, and in what parameter regime. With (10) promoted to a model parameter, a third control appears: *decorrelation* of the cross-location noise, which the inherited model held fixed at $\rho = 0$.

References

[1] D. M. Green and J. A. Swets. *Signal Detection Theory and Psychophysics*. Wiley, New York, 1966.

[2] N. A. Macmillan and C. D. Creelman. *Detection Theory: A User's Guide*. 2nd ed., Lawrence Erlbaum, 2005.

[3] Herman Lab. *When Does Value-Directed Attention Matter? A Normative Model with Independent Attentional Benefit and Cost*. Manuscript, 2026-04-09 (target of the rebuild).